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A water quality index—do we dare?*

By Robert M. Brown, Nina I. McClelland, Rolf A. Deininger & Ronald G. Tozer**

For more than twenty years, demands for billions of dollars of expenditures for water pollution control have been made on American municipalities and industries; but no provision has yet been made for keeping the public informed, in simple and understandable terms, as to what this effort and investment is achieving—or not achieving—in water quality enhancement. While these demands are entirely appropriate, it is the apotheosis of irony that a form of quality-explicit communication between professionals and the public has not been developed.

This year (1970) will go down in history as the "year of the environment." Probably never before have so many people become aware of the quality, fate and future of the environment. Evidence of environmental degradation is everywhere, and the problems of water pollution share the front pages of the newspapers with major political and economic events. Everybody talks about water quality, or even more about the lack of it; yet few people stop for a moment and think about how this is being measured. Federal, state and local agencies collect data on a myraid of individual parameters, which end up in voluminous files or occasionally get published in booklets —in a form which is hard to digest and assimilate. What is clearly needed is a yardstick of water quality which takes into consideration the most significant parameters.

In calling for a yardstick to measure water quality, a number of voices are projected. The need for developing a uniform method for measuring the results of water pollution control programs has long been recognized by environmental engineers, and expressed through their various professional organizations. For example, in January 1959 the Committee on National Water Policy of the Conference of State Sanitary Engineers (CSSE) proposed that an objective study be initiated to develop a uniform method for measuring water quality. As a result of this action, this Committee and the Conference of State and Interstate Water Pollution Control Administrators (CSIWPCA) were requested to explore the development of suggested criteria for measuring the progress of water pollution control programs. While study was initiated and action contemplated, nothing tangible emerged.

In their report dated November 1965, the Environmental Pollution Panel of the President's Science Advisory Committee[1] recommended "that the Federal Government stimulate development of a method for assigning a numerical index of chemical pollution to water samples." It was suggested that "the method should be sensitive to most chemical pollutants, and its result should be roughly proportional to the unfavorable effects of the pollution on man or aquatic life. Such an index will allow us to follow many important changes in general water quality." Again, there was no visible or applicable response.

As recently as January 1970, the Environmental Study Group of the National Academy of Science[2] called for the development and utilization of various environmental indices (including "water purity"—to be combined and weighted into an overall Environmental Quality Index "which could become a powerful tool in developing priorities among programs affecting the environment." This group recommended that comprehensive and systematic monitoring of environmental quality be given the highest priority. Also in January 1970, in a report to the Senate Committee on Public Works entitled "The Case for National Environmental Laboratories,"[3] the need for environmental quality indices was again noted as having the highest priority.

In response to these continued expressions of need, development of a Water Quality Index (WQI) was undertaken. This paper reports on the methodology and some of the preliminary findings of this effort.

Previous efforts and studies. The most notable research on this continent in this type of study is the work of Horton.[4] He defined a water quality index based entirely on chemical and physical measurements. Eight parameters—sewage treatment, DO, pH, coliform density, specific conductance, carbon chloroform extract, alkalinity and chlorides—were selected; rating scales were assigned; and each parameter was weighted according to its relative significance in overall stream quality. Two additional parameters, "temperature" and "obvious pollution", were not rated to show gradations in quality, but were handled as "yes or no" indicators in the index formulation. There is no indication that opinion research was carried out in defining these weights. All judgments were established by the author and his associates. The paper was aimed at calling attention to the usefulness of such a method, rather than establishing the validity of the individual weights and grades.

In Europe, Liebman[5] proposed a water quality index based on chemical and biological parameters, and proceeded to chart the water quality of the entire state of Bavaria (West Germany). His method of assigning weights and quality ratings to the individual parameters has not been widely accepted, since all of them were based on his individual opinion.

Applied methodology. In the formulation of a WQI many different methods can be used and postulated.

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immediate task is to decide which parameters to include, and to get a quality rating for each parameter.

In an attempt to alleviate the limitations of previous efforts to derive a WQI—due to the "subjective" establishment of selected parameters, rating scales and parameter weightings—systematic opinion research technology was utilized to incorporate the judgment of a large and diverse panel. The research procedure was designed to minimize formation of a judgment based on any professional viewpoint or individual geographical area.

The DELPHI method, developed by Rand Corporation,[6] is an opinion research technique which can be utilized to extract information from a group or respondents. The procedure used in formulating this WQI attempted to incorporate many aspects of the DELPHI process. It provided for anonymity of individual responses in the questioning of a panel of persons with expertise in water resource management, while enabling the panel to view the total judgment of all respondents—a positive characteristic of group interaction. The latter was achieved through controlled feedback of group response.

Procedure. A panel of 142 persons with expertise in water quality management was selected for this study, as indicated in Table 1. The respondents received a series of mailed questionnaires and feedback information.

In the first questionnaire, the respondents were asked to consider 35 parameters for possible inclusion in a water quality index. These parameters, listed in Table 2, were somewhat arbitrarily selected and arranged for initial consideration by the respondents. Opportunity was provided for the respondents to add to the list any parameters which they believed should be included in the WQI. Each parameter was to be designated according to one of the following categories: "do not include," "undecided" or "include." Respondents were asked to rate only those parameters marked "include," according to their significance as contributors to overall water quality. This rating was done on a scale of "1" (highest relative significance) to "5" (lowest relative significance). Table 3 shows the beginning of the questionnaire and a sample response. Of the total panel of 142 members, 102 respondents (72 percent) completed and returned the first questionnaire. However, of these, only 94 were returned in time to receive the second questionnaire.

The second mailing included a computer printout of the results from the first questionnaire, as shown in Figure 1. Respondents were instructed to note their individual responses for each parameter and compare them with those of the entire group. In view of this feedback information, respondents were then asked to review their original judgment and to modify their initial responses if they wished. The intent here was to initiate a greater convergence of opinion concerning how the various parameters rated with respect to their effect on overall water quality. (However, it should be noted that there was little change in the significance ratings expressed in Questionnaire No. 2 when compared with the initial round.)

Some additional parameters, i.e., chromium (hexavalent), total organic carbon, cyanides, conductivity, lead, arsenic, cadmium, selenium and zinc, which had been added to the first questionnaire by several respondents were also introduced for consideration by the group at this time. Finally, the respondents were asked to designate not more than 15 parameters, which they considered to be the "most important" for inclusion in a water quality index. The complete list of parameters was presented for them to choose from, arranged in decreasing order of significance as determined by the average rating of the entire group from Questionnaire No. 1. (See Table 4). Of the 94 respondents receiving this second questionnaire, 77 completed and returned it, an 82 percent response rate.

Table 1: Professional fields of invited panel participants

Regulatory officials (federal, interstate, state, territorial and regional)	101
Local public utilities managers	5
Consulting engineers	6
Academicians	26
Others (industrial waste control engineers and representatives of professional organizations)	4
Total	142

Table 2: Listing of parameters included in first questionnaire

Dissolved oxygen	Manganese
Biochemical oxygen demand (5-day)	Magnesium
	Nitrate and nitrite
Chemical oxygen demand	Phosphate
Dissolved solids	Silica
Color	Sodium and potassium
pH	Sulfate
Carbon chloroform extract	Temperature
Hardness	Pesticides
Alkalinity	Coliform organisms
Acidity	Radioactivity
Aluminum	Phenols
Ammonia	Oil and grease
Bicarbonate	Herbicides
Calicium	Turbidity
Chloride	Total solids
Copper	Fecal coliforms
Fluoride	
Iron	

Table 3: Sample response to first WQI questionnaire

Parameter	1 Do Not Include	2 Undecided	3 Include	4 Significance (Rate from "1" highest to "5" lowest)
Dissolved oxygen			X	1
BOD_5			X	5
COD	X			
Color		X		
Turbidity			X	3

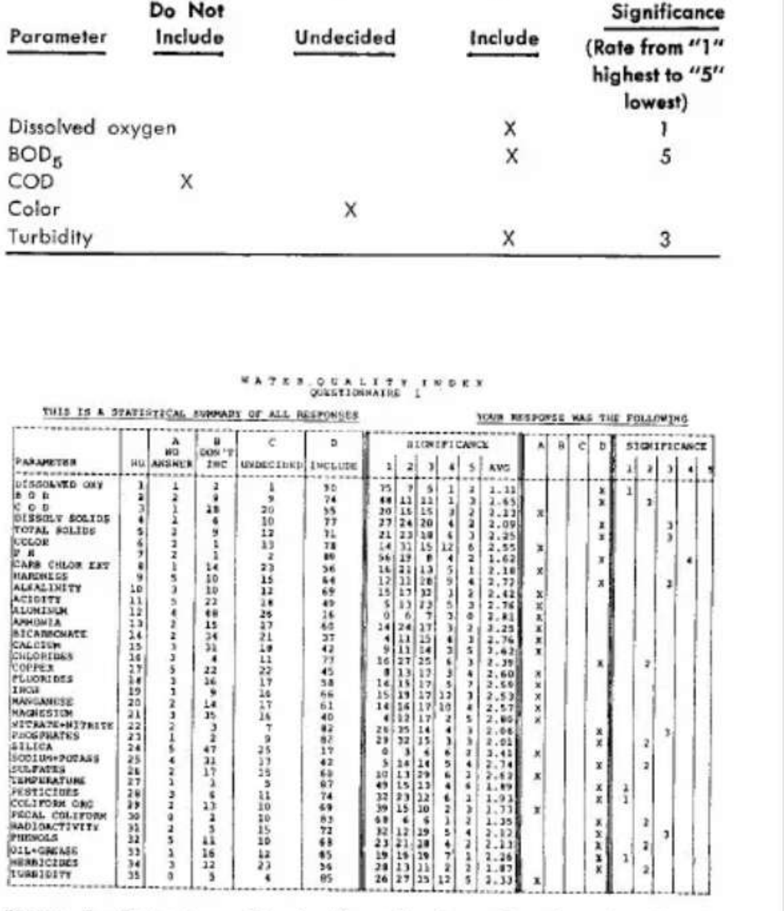
WATER QUALITY INDEX
QUESTIONNAIRE I
THIS IS A STATISTICAL SUMMARY OF ALL RESPONSES
YOUR RESPONSE WAS THE FOLLOWING

Figure 1: Computer printout of results from Qu.stionnaire No. 1.

Table 4: Parameters included in second WQI questionnaire and significance ratings from the first WQI questionnaire

Parameter	Average rated significance on scale of "1" (highest) to "5" (lowest)	Number of respondents indicating parameter should be included
Dissolved oxygen	1.31	90
Fecal coliforms	1.35	83
pH	1.62	89
Biochemical oxygen demand (5-day)	1.65	74
Coliform organisms	1.77	69
Herbicides	1.87	56
Temperature	1.89	87
Pesticides	1.93	74
Phosphates	2.01	82
Nitrates	2.06	82
Dissolved solids	2.09	77
Radioactivity	2.12	72
Phenols	2.13	68
Chemical oxygen demand	2.13	55
Carbon chloroform extract	2.18	56
Ammonia	2.25	60
Total solids	2.25	71
Oil and grease	2.26	65
Turbidity	2.33	85
Chlorides	2.39	77
Alkalinity	2.42	65
Iron	2.53	66
Color	2.55	78
Manganese	2.57	61
Fluorides	2.59	58
Copper	2.60	45
Sulfates	2.62	60
Calcium	2.62	42
Hardness	2.72	64
Sodium and potassium	2.74	42
Acidity	2.76	49
Bicarbonate	2.76	37
Magnesium	2.80	40
Aluminum	2.81	16
Silica	3.41	17

Extra parameters (added by one or more respondents to questionnaire no. 1)

Chromium (hexavalent)
Total organic carbon
Cyanides
Conductivity
Lead
Arsenic
Cadmium
Selenium
Zinc

Table 5: List of eleven most significant parameters

Dissolved oxygen
Biochemical oxygen demand (5-day)
Turbidity
Total solids
Nitrates
Phosphates
pH
Temperature
Fecal coliforms
Pesticides
Toxic elements

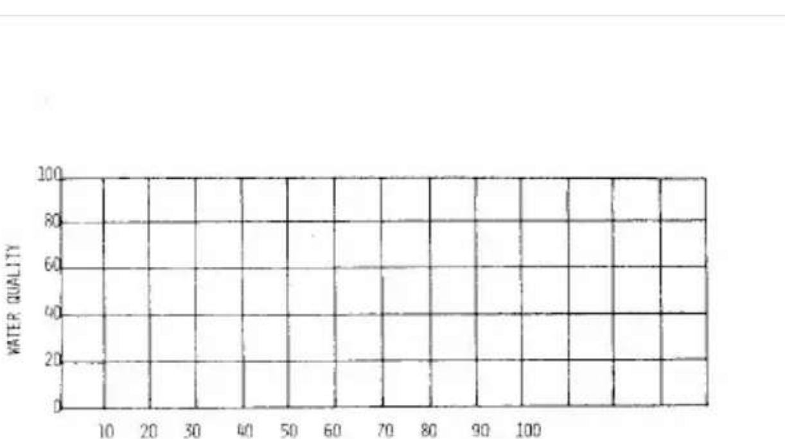

Figure 2: Graph for turbidity parameter.

Utilizing the expert opinion derived from the initial rounds of the study, a list of eleven parameters was established for further consideration. (See Table 5). In establishing this list, some parameters were grouped which had previously been presented separately.

In Questionnaire No. 3, respondents were asked to assign values for the variation in level of water quality produced by different strengths of nine of these selected parameters. This was accomplished by utilizing a series of graphs. On the vertical axis, levels of "water quality" from 0 to 100 were indicated, while various levels (or strengths) of the particular parameter were arranged along the horizontal axis. For example, the graph for "turbidity" is shown in Figure 2.

The respondents were asked to draw in a curve which, in their judgment, represented the variation in level of water quality produced by the various possible measurements of each respective parameter. This information was later used by combining the "judgments" of all respondents to produce a set of "average curves"—one for each parameter. Figure 3 shows the curves for "dissolved oxygen" and "turbidity." It is interesting to note the greater agreement on dissolved oxygen as evidenced by the much narrower band including 80 percent of the responses.

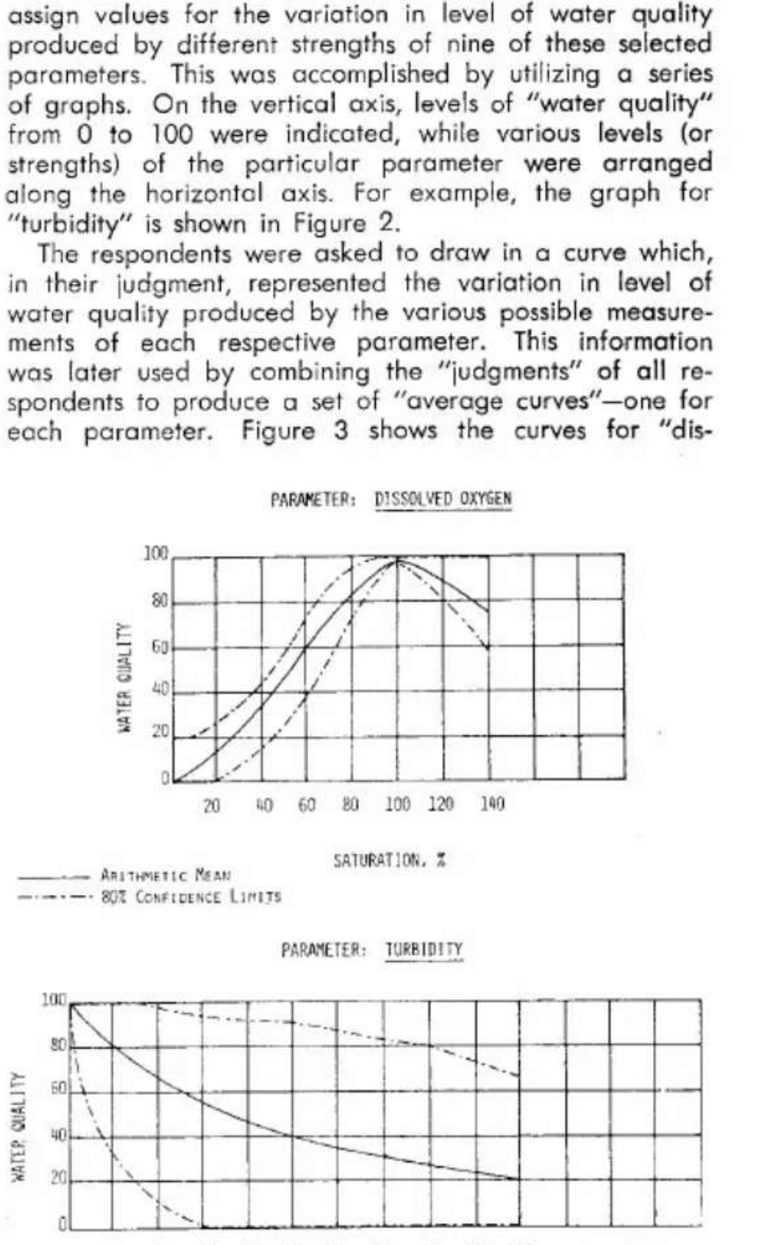

Figure 3: Average curves for two parameters.

Special procedures seemed necessary for "pesticides" and "toxic elements," and the respondents were asked to evaluate these. For pesticides it was proposed that "if the total content of detected pesticides (of all types) exceeds 0.1 mg/l (100 ppb), the water be automatically registered at zero, the lowest value on the water quality index scale." The suggested procedure for including "toxic elements" in the WQI "would be to set a critical upper limit for the presence of each of these elements. If any toxic element exceeds its assigned upper limit, the water quality index would automatically register as zero." The maximum permissible levels for toxic elements contained in the Public Health Service **Drinking Water Standards** of 1962[7] were proposed as those to be observed. The respondents were asked whether they agreed or disagreed with the suggested procedure for each of these parameters, and why. The results indicated that further study of these parameters will be necessary.

The third mailing also sought information which contributed to the formation of "parameter weightings" in a final numerical expression. Thus, respondents were asked to compare relative overall water quality, using a scale of "1" (highest relative value) to "5" (lowest relative value). This operation differed from the initial rating of significance (Questionaire No. 1) in that now only the final eleven parameters were being considered. From these data, weightings were derived for each of the parameters to comprise the final WQI, according to the following procedure:

1. (1) Arithmetic means were calculated for the significance ratings returned for all parameters except "pesticides" and "toxic elements." (See Table 6).
2. (2) To convert the ratings into weights, a "temporary weight" of 1.0 was assigned to the parameter which received the highest significance rating, i.e., dissolved oxygen.
3. (3) To preserve the ordering and relative ratios returned by the panel of respondents, all other temporary weights were obtained by dividing each individual mean rating into the highest.
4. (4) Each "temporary weight" was then divided by the sum of all weights to arrive at the final weights shown in Table 6. As indicated in the table, the final weight for dissolved oxygen is 0.17 compared with 0.08 for total solids. The ratio of 0.17 to 0.08 is equal to the inverse of the original ratings, i.e., 1.4 to 3.2. (Note: These ratios were only approximately preserved in rounding the final weights to two significant digits).

Table 6: Significance ratings and weights for nine parameters included in the WQI

Parameters	Mean of all significance ratings returned by respondents	Temporary weights	Final weights
Dissolved oxygen	1.4	1.0	0.17
Fecal coliform density	1.5	0.9	0.15
pH	2.1	0.7	0.12
Biochemical oxygen demand (5-day)	2.3	0.6	0.10
Nitrates	2.4	0.6	0.10
Phosphates	2.4	0.6	0.10
Temperature	2.4	0.6	0.10
Turbidity	2.9	0.5	0.08
Total solids	3.2	0.4	0.08
		Total = Σ =	1.00

Table 7: Best and worst possible WQI

Parameters	Measured values	Individual quality rating (q_i)	Weights (w_i)	Overall quality rating ($q_i \times w_i$)
DO, percent sat.	100	98	0.17	16.7
Fecal coliform density, lb/100 ml	0	100	0.15	15.0
pH	7.0	92	0.11	10.1
BOD_5, mg/l	0.0	100	0.11	11.0
Nitrate, mg/l	0.0	98	0.10	9.8
Phosphate, mg/l	0.0	98	0.10	9.8
Temperature °C departure from equil	0.0	94	0.10	9.4
Turbidity, units	0	98	0.08	7.8
Total solids, mg/l	25	84	0.08	6.7
				$WQI = \Sigma w_i q_i = 96.3$

Worst quality stream

Parameters	Measured values	Individual quality rating (q_i)	Weights (w_i)	Overall quality rating ($q_i \times w_i$)
DO, percent sat.	0	0	0.17	0
Fecal coliform density, lb/100 ml	5	4	0.15	0.6
pH	2	4	0.11	0.4
BOD_5, mg/l	30	8	0.11	0.9
Nitrate, mg/l	100	2	0.10	0.2
Phosphate, mg/l	10	6	0.10	0.6
Temperature °C departure from equil	+15	10	0.10	1.0
Turbidity, units	100	18	0.08	1.4
Total solids, mg/l	500	30	0.08	2.4
				$WQI = \Sigma w_i q_i = 7.5$

Index formulation: A weighted mean index was decided upon with the form:

$$WQI = \sum_{i=1}^{n} w_i q_i$$

where:

WQI = the water quality index, a number between 0 and 100.

q_i = the quality of the ith parameter, a number between 0 and 100.

w_i = the unit weight of the ith parameter, a number between 0 and 1; and

$$\sum_{i=1}^{n} w_i = 1$$

n = number of parameters

This type of index works well if all of the individual parameters are independent of each other.*

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*This is not necessarily the case for all parameters considered, e.g., temperature influences DO, etc.

Parameters	Measured values	quality rating (q_i)	Weights (w_i)	quality rating ($q_i \times w_i$)
DO, percent sat.	80.0	86	0.17	14.6
Fecal coliform density, lb/100 ml	10.0	68	0.15	10.2
pH	7.5	92	0.12	11.0
BOD_5, mg/l	2.0	75	0.10	7.5
Nitrate, mg/l	10.0	48	0.10	4.8
Phosphate, mg/l	1.0	40	0.10	4.0
Temperature °C. departure from quil	0.0	95	0.10	9.5
Turbidity, units	10.0	76	0.08	6.1
Total solids, mg/l	100.0	82	0.08	6.6
				$WQI = \Sigma w_i q_i = 74.3$

Actual operation of the WQI may be illustrated through the utilization of hypothetical data. Table 7 shows the "best" and "worst" WQI reading which could possibly occur. Table 8 represents a typical index value which might be derived from a stream of moderately high quality.

Future work. Seventy respondents completed and returned the third questionnaire, for a 74 percent response rate. Analysis of the results indicates the need for further refinement of the data. It is desirable to feed back this information to the respondents, and attempt to initiate greater convergence in their expert judgments. In addition, some breakdown of the responses by geographical area and professional background is contemplated. This project will be continued.

Comments and judgments of respondents throughout the study reflected a variety of concerns in deriving a water quality index. Perhaps the most frequently stated was the matter of rating overall water quality, rather than classifying water for a particular use (e.g., drinking water). Related to this difficulty was an apparent tendency for some respondents to be heavily influenced in their judgment of parameter suitability for inclusion in a WQI, by factors such as data availability and existing analytical methodologies for measuring the various parameters. It was concluded, however, that these consideration did not prevent the establishment of an index of general water quality.

In the future, an effort will be made to seek greater convergence in the parameter data. Further investigation of a single WQI expression which recognizes and incorporates specific water functions (e.g., agriculture, industry, fish and wildlife, recreation, esthetics and drinking water) is also contemplated.

A review of previously reported field studies has failed to provide information for all nine parameters included

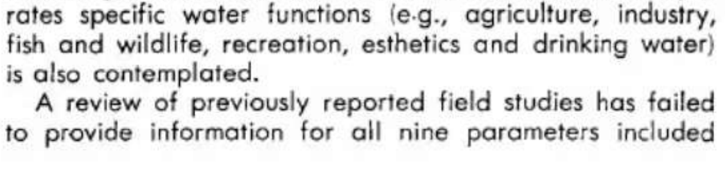

Figure 4: WQI values.

logically be compared with that from another. Thus, to refine and validate this technique for expressing overall water quality, field application of the WQI is proposed. In particular, regional application will be investigated.

A workable index will be a useful mechanism for indicating the progress of water quality improvement programs and for the comparison of water conditions over time and in various geographical locations. In this regard, the feasibility of using a color spectrum scale to illustrate water quality throughout an entire region or state is being explored. This concept is hypothetically illustrated in Figure 4, in which water quality index values are related to shades of colors in the spectrum. In this figure, the poorest quality is a dark red. A narrow band of yellow depicts water of medium quality with an index of 50, and the best quality is dark blue.

A variety of public communication and education usages for the index are envisioned. For example, monthly published indices for selected bodies of water would be a useful device for informing the public of pollution abatement progress by municipalities, industries, etc. The positive citizen awareness and action programs which have been initiated in a number of cities through the use of air pollution indices might well be duplicated for water quality.

Conclusion. It is concluded, based on experience and data derived from this study, that a single numerical expression which reflects the composite influence of significant parameters of water quality is feasible. The WQI provides a meaningful, uniform method for assessing the overall quality of a stream, comparing stream conditions at different points in space and time, and measuring progress — or lack of progress — in water pollution abatement programs. With the current rate of environmental degradation, improved procedures for ecological monitoring and environmental education must be developed. The WQI can perform an essential role in this process. A water quality index — do we dare? Most emphatically, yes!

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